Srijan Challapalli

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EDUCATION

Purdue University West Lafayette, IN
B.S. in Artificial Intelligence *Expected May 2028*
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Linear Algebra, Probability & Statistics, Discrete Math

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Swift, Rust, Solidity, HTML/CSS
AI & ML: XGBoost, Pandas, NumPy, LLM/agent orchestration, multi-model routing, n8n, LlamaParse, JSON Schema, REST APIs
Backend & Systems: Next.js, React, SwiftUI, Flask, Express, Supabase, PostgreSQL, HealthKit, Criterion, Linux perf, Git/GitHub
Web3: Solidity, Foundry, wagmi, EIP-712

EXPERIENCE

MergeWorks Remote
Software Engineer Intern *May 2026 – Present*
- Built an M&A due-diligence pipeline that turns messy financial packets (Excel, QuickBooks, tax returns, scanned PDFs) into source-cited, audit-ready analysis in ~71s per document at p50 and 125s at p95, replacing hours of manual review.
- Removed the LLM from every financial calculation by routing all arithmetic into a deterministic reconciliation sandbox of 5 balance checks (gross profit, EBITDA, equity, margin consistency, order-of-magnitude) at a 2% tolerance, keeping every reported figure auditable.
- Isolated faults across 90+ code-split React modules with paired Suspense and error boundaries, and added an idempotent intake guard that normalizes (project, filename, size) and returns HTTP 409 to block duplicate paid LLM runs.
- Led a zero-downtime dual-write migration from n8n Data Tables to Postgres, cut the read path over to Supabase, retired 6 legacy webhook workflows draining execution quota, and backed the app with 111 passing unit tests.

Purdue University · STyGIANet Lab West Lafayette, IN
Undergraduate Researcher *Aug. 2026 – Present*
- Researching GPU interconnects and collective communication for distributed AI systems, focusing on adaptive network topologies and programmable photonic interconnects that reduce communication bottlenecks in large-scale training.
- Studying performance tradeoffs among reconfiguration overhead, congestion, propagation delay, and communication-completion time to inform topology-reconfiguration decisions for large-scale GPU training and inference workloads.

Cosmos Granite & Marble Chantilly, VA
Software Engineer Intern *May 2025 – Aug. 2025*
- Built a self-contained Python data-collection tool (Google Maps API via SerpAPI) that surfaced 1,000+ competitor and prospect fabricators across target states, capturing name, phone, website, and ratings per record and replacing hours of manual prospecting.
- Engineered a contact-extraction pipeline that scrapes homepage and contact pages (mailto plus HTML email parsing with junk-domain filtering) and writes deduplicated, per-state datasets to Excel, eliminating repetitive manual list-building.

PROJECTS

ChitYap: *Swift, SwiftUI, Supabase, PostgreSQL, Apple Vision, HealthKit, APNs*
- Co-built a native SwiftUI iOS app on Supabase/Postgres with a 4-channel realtime fan-in (feed, notifications, friendships, requests) and foreground-reconnect handling, backed by ~180 SQL migrations, 17 feed-path indexes, and rate limiting.
- Built an on-device Apple Vision body-pose rep-counter running at 15–30 FPS with multi-frame confirmation and no network round-trip; cut row-level-security overhead across 60+ policies by hoisting auth.uid() to one evaluation per query.
- Architected the app in layered Views, Managers, Services, and 12 API modules enforced by architecture tests, with push notifications fanned out through Postgres SECURITY DEFINER RPCs and an offline post queue that retries failed writes from disk.

LSM-Tree Storage Engine: *Rust, Tokio, Criterion, Linux perf*
- Building a persistent key-value store in Rust (the LSM-tree design behind RocksDB/LevelDB): write-ahead log for durability, an in-memory memtable, flush to immutable sorted SSTables on disk, and bloom filters to skip disk reads on absent keys.
- Implementing a background compaction thread (Arc/Mutex + channels) that bounds read amplification and reclaims space, with throughput and p99 latency benchmarked via Criterion, plus crash recovery that replays the write-ahead log on startup.

Secretariat: *TypeScript, Solidity, Next.js, Foundry, XGBoost, Express · ETHDenver, Feb. 2026*
- Built a full-stack tokenized-asset marketplace at ETHDenver (Next.js/wagmi, 17 Solidity contracts); hit the EIP-170 24KB contract-size limit and split the vault via a factory/deployer pattern to land the core contract at 24,382 bytes, 194 under the cap.
- Wrote an off-chain XGBoost valuation engine in TypeScript (32 features) benchmarked at ~10 µs/prediction (~95K predictions/sec, single-threaded, zero native deps), covered by 48 passing Foundry tests at 52% line coverage on the core vault.
- Implemented EIP-712-signed multisig governance (submit, confirm, and execute proposal flow) and KYC-gated access via an accredited-investor registry, wiring an off-chain Express oracle to commit valuations on-chain through an ERC-4337 agent wallet.

INTERESTS

Badminton, Football, Weightlifting, Poker, Cliff Diving, Hiking, Video Games